👩🏽‍🏫 Staff Guide · Ag Science 1 · Unit 3 U.S. Food & Farm Facts (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Torretta's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “US Food and Farm Facts Notes + Quiz (10 questions), Food and Facts Pres/notes, The Future of Farming video, Biggest Little Farm, Agriculture Literacy Poster”)	**Level 2 piece**
US Food and Farm Facts Notes (25 pts) + Quiz (10 Qs)	Notes + modified quiz. L2 quiz = his 10 questions in his order with 3 choices; L1 = 8 of them with pictures (his question numbers on the key)
Food and Facts Pres/notes (20 pts)	No file in the export (check with Mr. Torretta); the slides cover the same Farm Bureau facts
The Future of Farming video (10 pts) · Biggest Little Farm (15 pts)	Video guides (Biggest Little Farm sheet: check with Mr. Torretta)
Agriculture Literacy Poster (30 pts)	Ag Literacy Poster packet
Class slides	Deck: 22 slides, Words to Know first, hold-up-1-2-3 checks, recap
Class notes (Blank Out)	Guided notes: 16 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	10 picture cards (cut apart)
—	Review game (match + order) on the class site

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Read the quiz aloud; allow pointing
- Use the notes during the quiz (his slide says “use these notes for the quiz”)

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Quiz: same 10 points; L1 answers 8 picture questions (scale to 10)
- **Notes**: same points as the class Blank Out notes when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with Mr. Torretta”.

6. Photo credits (open licenses)

Tractor in field - geograph.org.uk - 8060175.jpg — N Chadwick, CC BY-SA 2.0, Wikimedia Commons · Combine in wheat field.jpg — Dan Kollmann, CC BY-SA 3.0, Wikimedia Commons · Small greenhouse with grapevines escaping.jpg — W.carter, CC0, Wikimedia Commons · Main Dairy Barn CEF Ottawa.jpg — Matti Blume, CC BY-SA 4.0, Wikimedia Commons · Diagrams/pictures from Mr. Torretta's class slides: earth.png

👩‍🏫 Staff Guide · Ag Literacy Poster (Level 2)

Ag Science 1 · Unit 3 · goes with the class “Agriculture Literacy Poster Project” (30 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Project card**: Ag Literacy Contest: a poster that promotes agriculture and busts a myth (PosterMyWall, 8.5 × 11). 6 steps, one per box. **check with Mr. Torretta**: this year's PosterMyWall project name for your class (ask in class; it is not posted here)
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Which topic do you like?”
- “Is that fact true? Where did it come from?”
- “Can you read it from the door?”

3. Modifications

- Planner + 3 facts with sources
- Template allowed

4. What to grade (30 pts)

- Same 30 points; rubric as written with the planner as support
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Diagrams/pictures from Mr. Torretta's class slides: food-variety.png

🚜 U.S. Food and Farm Facts

Ag Science 1 · Unit 3 U.S. Food & Farm Facts · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Agriculture keeps improving

1. Farmers grow _____ food using fewer resources than ever.
2. Careful stewardship led to a nearly _____ % decline in cropland erosion since 1982.
3. Farm and ranch families are about _____ % of the U.S. population.
4. Each U.S. farmer feeds about _____ people.
5. U.S. farmers produce about 2.1 _____ pounds of food a year.

Word bank: **154** **2** **50** **more** **trillion**

B. Consumers and trade

6. The _____ pays the least for food of any country.
7. Almost all farms are owned by _____.
8. Farmers get only about _____ cents of each retail food dollar.
9. Circle the top 3 countries the U.S. exports to:

China | Canada | Mexico | Hong Kong | Italy

Word bank: **16** **families** **United States**

C. Production and the future

10. Biotechnology increases crop _____ and controls pests, weeds, and diseases.
11. Precision agriculture uses _____ and auto-steer.
12. Precision ag increases yields, lowers _____, and reduces chemical use.
13. The world population may grow to _____ billion by 2050.
14. Match the product to where it comes from in ag:

Crayons	sheep
Lumber	trees
Wool sweater	soybeans

Word bank: **9** **cost** **GPS** **yields**

Level 2 version of the gen ed “Food and Farm Facts Blank Out (American Farm Bureau) — use these notes for the quiz.”

📖 U.S. Food & Farm Facts: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.

👩‍🌾

Farmer

A person who grows crops or raises animals. Each U.S. farmer feeds about 154 people.



👪

Family farm

About 97% of U.S. farms are owned by families.

💵

Food dollar

Of each $1 spent on food, the farmer gets only about 16 cents.

🚢

Export

Top U.S. farm export buyers: China, Canada, and Mexico.



Precision ag

Using GPS, auto-steer, and sensors to raise yields, lower costs, and use fewer chemicals.

Biotechnology

Using cells and genes to improve crops: higher yields, fewer pests and diseases.

Vertical farm

Growing crops in stacked layers indoors.



Population

About 9–10 billion people by 2050, so farmers must grow much more food.

Stewardship

Caring for soil, water, and nature. Cropland erosion dropped nearly 50% since 1982.

Yield

The amount of crop grown on an area of land.

🎨 Ag Literacy Poster

Ag Science 1 · Unit 3 · Level 2



Name: _____

Ag Literacy Contest: a poster that promotes agriculture and busts a myth (PosterMyWall, 8.5 × 11) · check with Mr. Torretta.

🧺 What I need

- ☐ 💡 **1** topic from the menu
- ☐ 🔎 **2–3** true facts with sources
- ☐ 🖼️ **1** image that fits my message
- ☐ ✍️ **1** my name on the poster

🌱 Steps: check each box when you finish

1. 💡 Pick a topic: benefits, statistics, myths vs. facts, or regional ag. ☐
2. 🎯 Write your message: who is your audience? ☐
3. 🔎 Find 3 facts; write each source. ☐
4. 🎨 Design on PosterMyWall: title, image, facts, readable from far away. ☐
5. ✍️ Add your name and a credits line. ☐
6. 📤 Download and submit. ☐

🎨 Poster planner

Message

Topic: _____ Audience: _____

Purpose (circle): myth busting · sharing a story · call to action

Facts and sources

1. Fact: _____ Source: _____
2. Fact: _____ Source: _____
3. Fact: _____ Source: _____

Design check

☐ Readable from a distance ☐ Image fits the message ☐ Creative fonts ☐ Good use of space ☐ My name ☐ Credits

💡 *Fact-check sites: USDA and the Washington Agriculture Statistics Service. Example myth: “Farms are run by big corporations.” Fact: about 97% are family farms.*

🎨 Ag Literacy Poster · Prep Sheet

Ag Science 1 · Unit 3 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
💡 **Thinker**	Picks the topic and message.	
🔎 **Fact checker**	Finds true facts and writes the source.	
🎨 **Designer**	Makes it big and easy to read.	
📤 **Submitter**	Downloads and turns it in.	

🧰 Tools and supplies: check when you have it

☐ 💻 **Chromebook** PosterMyWall (student login, no account)

☐ 🔎 **Fact sites** USDA, WA Ag Statistics

☐ 📝 **Planner** in this packet

☐ 🖨️ **Printer** 8.5 × 11

⚠️ Safety and manners focus

🔐 Use the class project name. Do not make an account or share personal info.

✅ Only use true facts and say where they came from.

🖼️ Use pictures you are allowed to use.

🎨 Ag Literacy Poster · Reflection

Ag Science 1 · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Poster draft ☐ Final poster

✅ Skills I used today (check all you did)

☐ 💡 I picked a topic and a message

☐ 🔎 I used true facts with sources

☐ 🎨 My poster is readable and neat

☐ 📤 I downloaded and submitted it

⭐ What myth or message did you pick? Why?

1. _____
2. _____

Ideas: **family farms** **food safety** **Whatcom ag**

🔧 What would you change?

1. _____
2. _____

Ideas: **bigger title** **better image** **another fact**

✅ U.S. Food & Farm Facts Quiz

Ag Science 1 · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 📈 Farmers grow ___ food using fewer resources than ever.

A. Less | B. More | C. About the same amount of

2. 💵 Which country pays the least for food?

A. Canada | B. Philippines | C. United States

3. 🌱 Stewardship has led to a nearly ___ decline in cropland erosion since 1982.

A. 50% | B. 20% | C. 90%

4. 👩‍🌾 Each U.S. farmer feeds about ___ people.

A. 10 | B. 50 | C. 154

5. ⚖️ U.S. farmers produce about ___ pounds of food a year.

A. 2.1 trillion | B. 1 million | C. 10 million

6. 👪 Almost all farms are owned by…

A. Corporations | B. Families | C. The government

7. 🚢 The top 3 countries the U.S. exports to are…

A. China, Canada, and Hong Kong | B. China, Canada, and Mexico | C. Hong Kong, Indonesia, and the Philippines

8. 💵 Farmers get only about 16 cents of every retail food dollar.

A. True | B. False | C. Only in Washington

9. 📡 Precision ag increases yields, lowers ___, and reduces chemical use.

A. Production | B. Yields | C. Cost

10. 🌍 The world population may grow to 9 billion by 2050.

A. True | B. False | C. It will shrink

🎬 The Biggest Little Farm (film)

Ag Science 1 · Unit 3 · Level 2 video guide (modified). Film shown in class (documentary about Apricot Lane Farms, California).

Name: _____

While you watch: circle the best answer

1. John and Molly start a farm that…

A. Grows only one crop | B. Works with nature (many plants and animals) | C. Uses no water

2. How did they fix the snail problem?

A. They moved away | B. They sold the trees | C. Ducks ate the snails

3. Owls and hawks helped control…

A. Gophers | B. Ducks | C. Bees

4. Biodiversity means…

A. Only cows | B. No animals | C. Many kinds of plants and animals

After you watch: finish the sentence

5. One problem the farm had:
_____ and how they solved it: _____ _____

6. Would you want to live on this farm? Why?
_____ _____

Goes with the class “Biggest little farm” (15 pts). Same points when complete.

🎬 The Future of Farming

Ag Science 1 · Unit 3 · Level 2 video guide (modified). YouTube: youtube.com/watch?v=Qmla9NLFBvU

Name: _____

While you watch: circle the best answer

1. With about 10 billion people by 2050, farms will need to ___ food production.

A. Stop | B. Double (greatly increase) | C. Cut in half

2. What are vertical farms?

A. Farms on hills | B. Tall corn | C. Crops grown in stacked layers indoors

3. Cows may be tracked with fit-…

A. bits (fitness trackers) | B. balls | C. boots

4. Farmed fish now passes ___ in consumption.

A. Water | B. Apples | C. Beef

5. AQUAmax and Artesian are ___-tolerant corn.

A. Drought | B. Snow | C. Salt

After you watch: finish the sentence

6. Would you eat synthetic (lab) meat? Why?

Yes / No because _____ _____

7. One idea to grow more food:

_____ _____

Goes with the class “The Future of Farming Video” (10 pts). Same points when complete.

🚜 U.S. Food and Farm Facts · ANSWER KEY

Ag Science 1 · Unit 3 U.S. Food & Farm Facts · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Agriculture keeps improving

1. Farmers grow **more** food using fewer resources than ever.
2. Careful stewardship led to a nearly **50** % decline in cropland erosion since 1982.
3. Farm and ranch families are about **2** % of the U.S. population.
4. Each U.S. farmer feeds about **154** people.
5. U.S. farmers produce about 2.1 **trillion** pounds of food a year.

Word bank: 154 · 2 · 50 · more · trillion

B. Consumers and trade

6. The **United States** pays the least for food of any country.
7. Almost all farms are owned by **families**.
8. Farmers get only about **16** cents of each retail food dollar.
9. Circle the top 3 countries the U.S. exports to:

China · **Canada** · **Mexico** · Hong Kong · Italy

Word bank: 16 · families · United States

C. Production and the future

10. Biotechnology increases crop **yields** and controls pests, weeds, and diseases.
11. Precision agriculture uses **GPS** and auto-steer.
12. Precision ag increases yields, lowers **cost**, and reduces chemical use.
13. The world population may grow to **9** billion by 2050.
14. Match the product to where it comes from in ag:

(Key: Crayons → soybeans · Lumber → trees · Wool sweater → sheep)

Crayons	sheep
Lumber	trees
Wool sweater	soybeans

Word bank: 9 · cost · GPS · yields

Level 2 version of the gen ed “Food and Farm Facts Blank Out (American Farm Bureau) — use these notes for the quiz.”

✅ U.S. Food & Farm Facts Quiz · ANSWER KEY

Ag Science 1 · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 📈 Farmers grow ___ food using fewer resources than ever. Gen ed #1

A. Less | **B. More** | C. About the same amount of

2. 💵 Which country pays the least for food? Gen ed #2

A. Canada | B. Philippines | **C. United States**

3. 🌱 Stewardship has led to a nearly ___ decline in cropland erosion since 1982. Gen ed #3

A. 50% | B. 20% | C. 90%

4. 👩‍🌾 Each U.S. farmer feeds about ___ people. Gen ed #4

A. 10 | B. 50 | **C. 154**

5. ⚖️ U.S. farmers produce about ___ pounds of food a year. Gen ed #5

A. 2.1 trillion | B. 1 million | C. 10 million

6. 👪 Almost all farms are owned by… Gen ed #6

A. Corporations | **B. Families** | C. The government

7. 🚢 The top 3 countries the U.S. exports to are… Gen ed #7

A. China, Canada, and Hong Kong | **B. China, Canada, and Mexico** | C. Hong Kong, Indonesia, and the Philippines

8. 💵 Farmers get only about 16 cents of every retail food dollar. Gen ed #8

A. True | B. False | C. Only in Washington

9. 📡 Precision ag increases yields, lowers ___, and reduces chemical use. Gen ed #9

A. Production | B. Yields | **C. Cost**

10. 🌍 The world population may grow to 9 billion by 2050. Gen ed #10

A. True | B. False | C. It will shrink

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “US Food & Farm Facts quiz (his 10 questions, same order, 3 choices)”. Score: __ / 10.

🎬 The Biggest Little Farm (film) · ANSWER KEY

Ag Science 1 · Unit 3 · Level 2 video guide (modified). Film shown in class (documentary about Apricot Lane Farms, California).

Name: _____

While you watch: circle the best answer

1. John and Molly start a farm that…

A. Grows only one crop

B. Works with nature (many plants and animals)

C. Uses no water

2. How did they fix the snail problem?

A. They moved away

B. They sold the trees

C. Ducks ate the snails

3. Owls and hawks helped control…

A. Gophers

B. Ducks

C. Bees

4. Biodiversity means…

A. Only cows

B. No animals

C. Many kinds of plants and animals

After you watch: finish the sentence

5. One problem the farm had:

_____ and how they solved it: _____ _____

6. Would you want to live on this farm? Why?

_____ _____

Goes with the class "Biggest little farm" (15 pts). Same points when complete. No question sheet for this film was in the export; check with Mr. Torretta for his questions.

🎬 The Future of Farming · ANSWER KEY

Ag Science 1 · Unit 3 · Level 2 video guide (modified). YouTube: youtube.com/watch?v=Qmla9NLFBvU

Name: _____

While you watch: circle the best answer

1. With about 10 billion people by 2050, farms will need to ___ food production.

A. Stop | **B. Double (greatly increase)** | C. Cut in half

2. What are vertical farms?

A. Farms on hills | B. Tall corn | **C. Crops grown in stacked layers indoors**

3. Cows may be tracked with fit-…

A. bits (fitness trackers) | B. balls | C. boots

4. Farmed fish now passes ___ in consumption.

A. Water | B. Apples | **C. Beef**

5. AQUAmax and Artesian are ___-tolerant corn.

A. Drought | B. Snow | C. Salt

After you watch: finish the sentence

6. Would you eat synthetic (lab) meat? Why?

Yes / No because _____ _____

7. One idea to grow more food:

_____ _____

Goes with the class “The Future of Farming Video” (10 pts). Same points when complete. Two causes of past productivity, the % lost to pests, and the rice blank are video-specific; check with Mr. Torretta.